

Polynomial Multiplication & Division

Algebra 1 (US Common Core) | Worksheet 1

1 Polynomial Multiplication

Definition

Polynomial multiplication involves applying the distributive property to multiply terms, combining like terms, and applying exponent rules ($x^a \cdot x^b = x^{a+b}$).

Worked Examples

$$\begin{aligned}2x(3x + 4) &= (2x \cdot 3x) + (2x \cdot 4) = 6x^2 + 8x \\(x + 3)(x - 5) &= x^2 - 5x + 3x - 15 = x^2 - 2x - 15\end{aligned}$$

Practice Problems

1. $3x(2x + 5)$
2. $-4y(3y - 2)$
3. $(x + 4)(x + 6)$
4. $(x - 3)(x + 7)$
5. $(2x + 1)(x - 4)$
6. $(3x - 2)(2x + 5)$
7. $(x + 5)^2$
8. $(2a - 3)(2a + 3)$
9. $(x + 2)(x^2 - 2x + 4)$
10. $(x - 1)(x^2 + 3x - 2)$
11. $2x(x + 3)(x - 1)$
12. $(3x^2 - 4)(2x^2 + 1)$
13. $(x - 4)^2$
14. $(5x - 2y)(x + 3y)$

2 Word Problems & Graphical Models

Definition

Algebraic word problems translate real-world scenarios into equations. Functions can be modeled visually using Cartesian coordinate graphs.

Worked Example

A car rental costs a flat fee of \$20 plus \$5 per hour ($y = 5x + 20$). How much does it cost for 4 hours?

$$y = 5(4) + 20 = 20 + 20 = \$40.$$

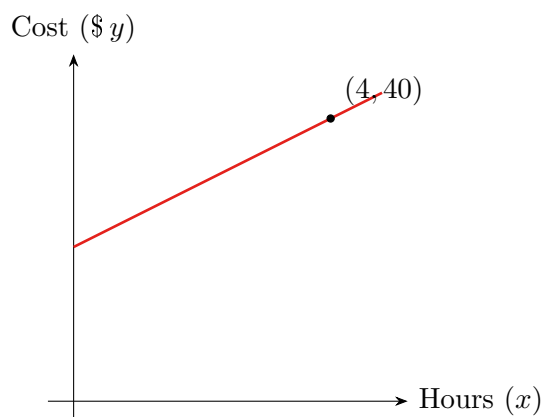


Figure 1: Rental Cost Model

Practice Problems

15. A rectangle has length $x + 5$ and width $x - 2$. Find its area expression.
16. The total cost C of buying x tickets is $C = 15x + 10$. Find C when $x = 6$.
17. A gardener doubles a square plot's side length s . Express the new area.
18. A line passes through $(0, 2)$ with slope $m = 3$. Write its slope-intercept equation.
19. Find the area of a right triangle with base $(x + 4)$ and height $6x$.
20. The sum of two consecutive integers is 47. Find the two integers.
21. A gym charges \$30 upfront plus \$15/month. Express cost for m months.
22. A pool fills at $y = 20x + 100$ gallons after x minutes. Find the volume at $x = 10$.
23. A linear graph intersects the y -axis at $(0, -4)$ and has slope $m = 2$. State its equation.
24. Express the volume of a cube with side length $(x + 2)$.
25. Two cars leave a town in opposite directions at 50 mph and 60 mph. How far apart are they after t hours?
26. A border of width 2 is added around a square of side x . Express the total area.
27. Find the x -intercept of the linear relation $y = 4x - 12$.

3 Polynomial Division (Remainder Zero)

Definition

Polynomial division with zero remainder means the divisor is an exact factor of the dividend: $\frac{P(x)}{D(x)} = Q(x)$ such that $P(x) = D(x) \cdot Q(x)$.

Worked Examples

$$\frac{12x^3 - 8x^2}{4x} = \frac{12x^3}{4x} - \frac{8x^2}{4x} = 3x^2 - 2x$$
$$\frac{x^2 + 5x + 6}{x + 2} = \frac{(x + 2)(x + 3)}{x + 2} = x + 3$$

Practice Problems

28. Divide $\frac{15x^2 + 10x}{5x}$

29. Divide $\frac{18a^4 - 12a^2}{6a}$

30. Divide $\frac{x^2 + 7x + 12}{x + 3}$

31. Divide $\frac{x^2 - 9}{x - 3}$

32. Divide $\frac{x^2 - 2x - 15}{x - 5}$

33. Divide $\frac{2x^2 + 7x + 3}{x + 3}$

34. Divide $\frac{x^2 - 16}{x + 4}$

35. Divide $\frac{3x^2 + 11x + 6}{3x + 2}$

36. Divide $\frac{x^3 + 4x^2 + 4x}{x}$

37. Divide $\frac{x^2 - 8x + 12}{x - 6}$

38. Divide $\frac{4x^2 - 25}{2x - 5}$

39. Divide $\frac{x^2 + 10x + 25}{x + 5}$

40. Divide $\frac{6x^3 - 9x^2 + 3x}{3x}$

4 Answer Key

1. $6x^2 + 15x$
2. $-12y^2 + 8y$
3. $x^2 + 10x + 24$
4. $x^2 + 4x - 21$
5. $2x^2 - 7x - 4$
6. $6x^2 + 11x - 10$
7. $x^2 + 10x + 25$
8. $4a^2 - 9$
9. $x^3 + 8$
10. $x^3 + 2x^2 - 5x + 2$
11. $2x^3 + 4x^2 - 6x$
12. $6x^4 - 5x^2 - 4$
13. $x^2 - 8x + 16$
14. $5x^2 + 13xy - 6y^2$
15. Area = $x^2 + 3x - 10$
16. \$100
17. Area = $4s^2$
18. $y = 3x + 2$
19. Area = $3x^2 + 12x$
20. 23 and 24
21. $C = 15m + 30$
22. 300 gallons
23. $y = 2x - 4$
24. Volume = $x^3 + 6x^2 + 12x + 8$
25. Distance = $110t$
26. Area = $(x + 4)^2 = x^2 + 8x + 16$
27. $(3, 0)$
28. $3x + 2$
29. $3a^3 - 2a$
30. $x + 4$
31. $x + 3$
32. $x + 3$
33. $2x + 1$
34. $x - 4$
35. $x + 3$
36. $x^2 + 4x + 4$
37. $x - 2$
38. $2x + 5$
39. $x + 5$
40. $2x^2 - 3x + 1$